# Finding causes

*How do we move from describing a problem to explaining why it exists?*



# Today

By the end of today, you will have

**Built a problem tree for your case that separates symptoms from proximate and structural causes, and tested a Five Whys chain against it.**

- Separate symptoms, proximate causes and structural causes.
- State the mechanism behind each causal link.
- Use Five Whys and the fishbone to generate candidate causes without treating them as proof.
- Distinguish correlation, causation and mechanism.

**Competencies: C1 · C2 · C5**

Session plan

| | |
|---|---|
| 10 min | **Opening problem** |
| 20 min | **Concept: levels of cause** |
| 15 min | **Demonstration: Madhyapur problem tree** |
| 45 min | **Thursday challenge: root cause investigation** |
| 20 min | **Own case: problem tree** |
| 10 min | **Debrief and exit task** |

# Opening problem

- **Children are not learning to read because teachers are absent. Is that a cause, a symptom, or a guess?**

Discuss with your neighbour for two minutes. What would you need to know before you believed it?

# Three levels of explanation

1

## Symptoms

What we see. Madhyapur: overflowing collection points and a near-full landfill.

2

## Proximate causes

The immediate reasons. Madhyapur: separated bags tipped into one compartment; households not separating.

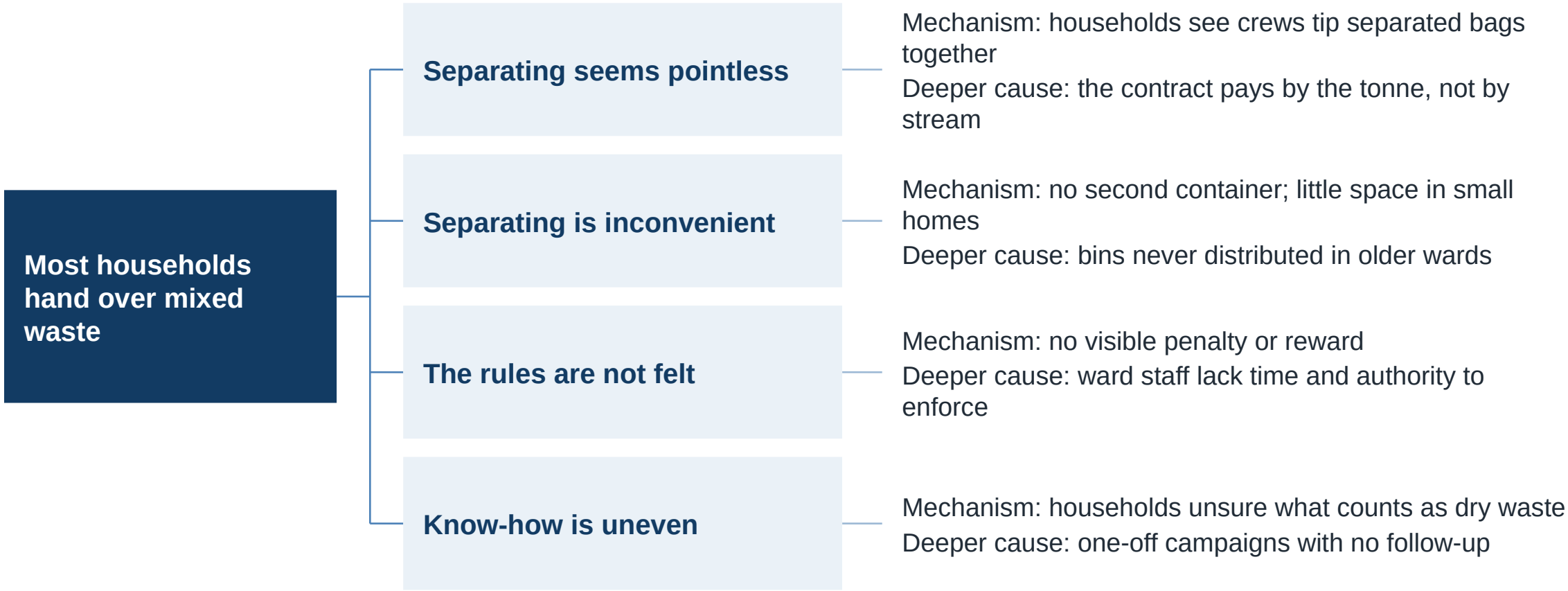
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## Structural causes

The conditions that produce the proximate causes. Madhyapur: a contract that pays by weight; no dry-waste processing.

A recommendation that treats only symptoms will need repeating every year.

# A problem tree for Madhyapur: causes and mechanisms




Effects sit above the core problem: a near-full landfill, lost recyclables, unsafe work for sanitation workers.

# Five Whys, used carefully

1

## Why do households mix waste?

They think it gets mixed anyway.

2

## Why do they think so?

They have seen crews tip bags together.

3

## Why do crews do it?

One compartment, and no time to separate.

4

## Why no compartment or time?

The contract pays by weight, not by stream.

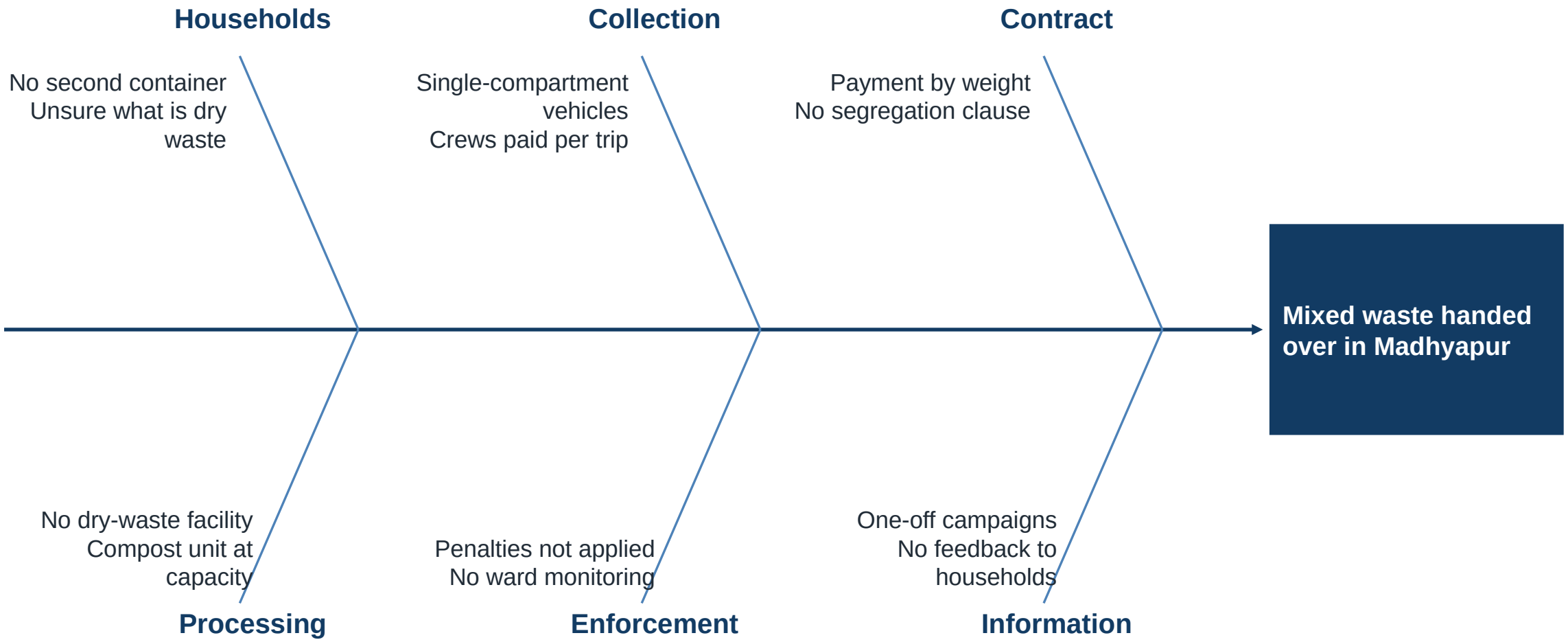
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## Why that contract?

It was written before the 2016 rules.

Each answer is a hypothesis. The chain is one path through the problem tree, not proof. Other paths may matter as much.

# A fishbone generates candidates across categories



# Root-cause tools: where they mislead

## Common misuse

- ○ Stopping at the first plausible chain
- ○ Ending at a cause you already believed
- ○ Treating a brainstorm as evidence
- ○ Ending with people do not care

## Better practice

- ● Run two or three chains from different starting points
- ● Check each link against the problem tree
- ● Mark every cause as a hypothesis in the Ledger
- ● Ask what would make caring rational

**Root-cause tools generate candidates. Evidence, in Week 3, decides between them.**

# Correlation, causation and mechanism

| Claim type | Example | What it shows |
|---|---|---|
| **Correlation** | Wards with fewer bins have lower separation | The two move together; nothing more |
| **Causal claim** | Fewer bins cause lower separation | Needs a mechanism and alternatives ruled out |
| **Mechanism** | Without a second container, small homes have nowhere to keep dry waste | Explains how the cause would work |
| **Alternative** | Older wards have both fewer bins and different housing | A third factor may drive both |

When you cite an association, say whether it supports a causal claim and why (competency C5).

# Activity: Root cause investigation

Separate symptoms, proximate causes and structural causes in an unfamiliar case, and defend the links.

### Scenario

A stylised district: many children cannot read a simple paragraph by the end of Grade 3. Eight cards name candidate causes, from teacher vacancies and multigrade classes to home language, harvest-time absence, syllabus-driven assessment and teaching at grade level rather than at the child's level.

### What to do

1. Sort the eight cards into symptoms, proximate causes and structural causes.
2. Build a problem tree, writing the mechanism on each arrow.
3. Run one Five Whys chain from the card you trust least.
4. Present your tree in two minutes; another pod challenges one link.
5. Mark which links you would test first, and why.

**Time** 45 minutes (Thursday challenge)

**Groups** Pods of four

**Output** A problem tree with mechanisms and one challenged link

# Your case: Worksheet 2.2

## Problem tree for your case

1. Write your core problem in the centre of the tree.
2. Place effects above and causes below; separate proximate from structural causes.
3. Write the mechanism on every link.
4. Run one Five Whys chain and check each answer against your tree.
5. Tag every cause as a hypothesis in your Ledger.

**Hand in**

Worksheet 2.2 by the start of Day 5, for inclusion in Gate 1.

**Quality check**

Symptoms, proximate and structural causes are separated.

Tag every claim in your Analyst's Ledger: fact, inference, assumption or hypothesis.

# What to take away

1. Separate what you see from why it happens, and immediate causes from structural ones.
2. Every causal arrow needs a mechanism.
3. Five Whys and fishbones generate candidates; they do not prove anything.
4. Most public problems have several causes. Single-cause stories are usually incomplete.

**Exit task** Write down the causal link in your tree that you are least sure of.

# Before the next session

### Independent work

Revise your problem tree after the challenge. For your two most important causal links, write what evidence would show the link to be wrong. About 45 minutes.

### Read

Andrews, Pritchett and Woolcock, Building State Capability (2017), the chapter on problem deconstruction, which uses the fishbone diagram (open access).

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### Next

## Day 5: Systems and scope lock

Tomorrow we look for the feedback loops that keep your problem in place, and lock your scope at Gate 1.